CSE 4345: Big Data Analytics

Week 10, Part 1 — Big Data in Real Applications: Social Media, Time Series & Health Data

Department of Computer Science & Engineering

Premier University, Chittagong

Semester 2026

Course & Lecture Information

Lecture Identity

Course:	CSE 4345
Week / Part:	10 of 11 | Part 1 of 2
CLO:	CLO4
PLO:	PLO(f)

CLO4

"Utilise the knowledge gained to solve big data problems and build a project applying big data tools and frameworks to real-world datasets"

Course Progression

Weeks 1–9: Foundations, storage, compute, query, pipelines, visualisation

Part 1 Covers

- Three application domains: social media, time series, healthcare
- **Domain A:** Social media data characteristics; community detection (Louvain); influence maximisation; sentiment analysis (VADER); trend detection
- **Domain B:** Time series properties; ARIMA; Facebook Prophet (trend + seasonality + holidays); LSTM; anomaly detection (STL, Isolation Forest); vectorised batch forecasting in Spark
- **Domain C:** Healthcare data sources (EHR, genomics, wearables, DICOM); clinical prediction models; privacy (HIPAA, PDPA Bangladesh); AlphaFold

Lecture Agenda

Domain Overview: Theory Meets Practice

Three Application Domains of Big Data

Social Media Analytics

- Graph-structured
- NLP / sentiment
- Community detect.
- Influence spread
- Trend detection

Tools: GraphX, NetworkXNeo4j, Twitter API v2

Time Series Analysis

- Ordered, temporal
- Seasonal, noisy
- ARIMA
- Prophet (Facebook)
- LSTM, anomaly det.

Tools: Prophet, SparkMLlib, statsmodels

Healthcare Big Data

- EHR records
- Genomics (omics)
- Wearables (IoT)
- Medical imaging
- Privacy critical

Tools: FHIR, TensorFlowHAPI, Spark MLlib

Shared Stack: HDFS / S3 | Kafka | Spark | Hive | Grafana / Superset

Domain A: Social Media Analytics

Social Media Data: Characteristics and Scale

Four Data Dimensions of Social Media

1. **Graph-structured:** Platforms are networks. Twitter/X has 41.7 million users (Kwak et al., 2010); Facebook has 3 billion. The graph (who follows whom, who retweets whom) is the primary analytical object — not the individual post.
2. **Temporal:** Every post, like, and share carries a timestamp. Events cascade through the network over minutes to hours. Time is a first-class dimension.
3. **Text (NLP):** Post content is unstructured natural language — short, noisy, abbreviation-heavy, and multilingual. Standard NLP pipelines must handle slang, emojis, and code-switching (Bangla–English mixing).

Scale Numbers (2024)

Platform	Posts/day
Twitter / X	500 M tweets
Facebook	100 B interactions
Instagram	100 M photos
TikTok	1 B videos viewed
YouTube	500 hours uploaded/min

Bangladesh context:

Facebook is the dominant social platform (50+ million users). During national disasters (floods, cyclones), crisis information spreads via Facebook Pages before official government channels.

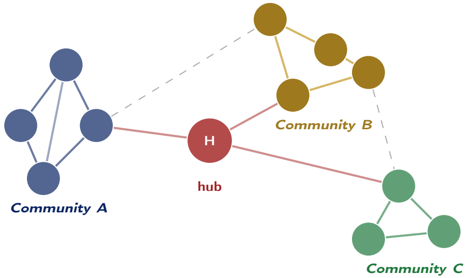
Community Detection and Influence Maximisation

Community Detection: Louvain Algorithm

A **community** in a social graph is a group of nodes more densely connected to each other than to the rest of the graph. Detecting communities reveals interest groups, political factions, professional clusters, or coordinated bot networks. **Louvain algorithm** (Blondel et al., 2008):

1. Start: each node is its own community
2. Greedily merge each node into the neighbour community that maximises **modularity** (fraction of edges within communities minus expected fraction by chance)
3. Aggregate merged communities into super-nodes; repeat
4. Terminate when modularity gain is zero

Community Structure in a Social Graph



Influence Maximisation

Which k **nodes to seed** so that information (or a product promotion) spreads to the maximum number of users? Kempe et al. (KDD 2003 Best Paper) proved this is NP-hard but gave a greedy

Sentiment Analysis and Trend Detection

Sentiment Analysis: VADER

VADER (Valence Aware Dictionary for Sentiment Reasoning, Hutto & Gilbert 2014) is a lexicon-and-rule-based sentiment analyser tuned specifically for **social media text**. It handles:

- Capitalisation: `GREAT` > `great`
- Punctuation: `great!!!` > `great`
- Degree modifiers: `very good`, `kind of bad`
- Negation: `not good` → negative
- Emojis: `:)` → positive score

Output: compound score $\in [-1, +1]$; positive (> 0.05), neutral, negative (< -0.05). **Spark pipeline:** Apply VADER to each tweet via a UDF on a DataFrame. Aggregate sentiment scores by time window and topic. Push to Grafana for the streaming

Trend Detection

Twitter's trending topics algorithm uses **Locality-Sensitive Hashing (LSH)** on character *n*-grams to detect sudden spikes in the frequency of a phrase *relative to its historical baseline*. A phrase that normally appears 10 times/hour and suddenly appears 10,000 times/hour is trending — regardless of absolute volume.

Bangladesh application: During Eid-ul-Adha 2024, the hashtag `#QurbaniEid` showed a 1,200× spike on X (Twitter) over 6 hours. A sentiment breakdown using VADER and a Bangla lexicon revealed 92% positive sentiment, confirming organic celebration rather than coordinated activity.

Ethical Risk

Domain B: Time Series Analysis

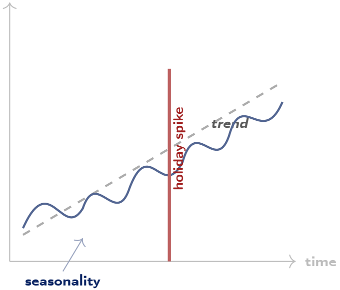
Time Series: Properties and Classical Forecasting (ARIMA)

What Makes Time Series Special

Time series data is fundamentally different from i.i.d. tabular data:

- **Order matters:** shuffling rows destroys the signal. Timestamps must be preserved.
- **Autocorrelation:** today's value is correlated with yesterday's. Standard regression that assumes independent observations is invalid.
- **Seasonality:** regular cyclical patterns (daily, weekly, annual) must be modelled explicitly.
- **Trend:** long-run upward or downward drift independent of seasonality.
- **Non-stationarity:** mean and variance may change over time. Most models require stationary input (achieved via differencing).

Typical Business Time Series



ARIMA Limitations

ARIMA assumes **stationarity**, struggles with complex multi-seasonal patterns, and requires expert parameter selection (p, d, q). At big data scale (10 000 time series), it cannot be manually

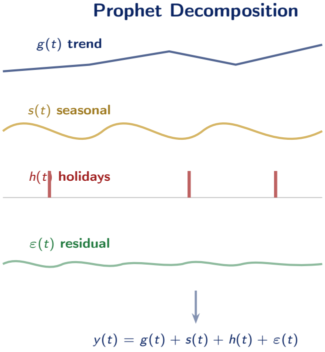
Facebook Prophet: Decomposable Forecasting at Scale

Prophet's Additive Model (Taylor & Letham, 2018)

Prophet decomposes a time series into interpretable components:

$$y(t) = g(t) + s(t) + h(t) + \varepsilon(t)$$

- $g(t)$: **trend** — piecewise linear or logistic growth; automatically detects **change points** where the trend shifts
- $s(t)$: **seasonality** — Fourier series capturing annual, weekly, and daily cycles; handles multiple simultaneous seasonalities
- $h(t)$: **holidays and events** — user-specified list of irregular one-off events (Eid, Puja, election



Anomaly Detection and Spark Batch Forecasting

Anomaly Detection in Time Series

An **anomaly** (outlier, change point, or contextual deviation) is a data point that does not conform to the expected pattern. Two methods:

1. **STL Decomposition** (Seasonal-Trend decomposition using LOESS): Decompose the series into trend, seasonal, and residual. Flag residuals whose absolute value exceeds $k \times$ IQR of residuals as anomalies. Interpretable; works well for regular seasonality.
2. **Isolation Forest:** Ensemble of random trees that *isolates* anomalies by partitioning the feature space. An anomaly requires fewer splits to isolate $\Rightarrow$ shorter path length. No assumption about data distribution; scales to millions of observations in Spark MLlib

Vectorised Batch Forecasting in Spark

Problem: A retail chain needs 90-day demand forecasts for 5,000 store–product SKU combinations. Fitting 5,000 separate Prophet models sequentially on one machine takes 8+ hours. **Spark solution:**

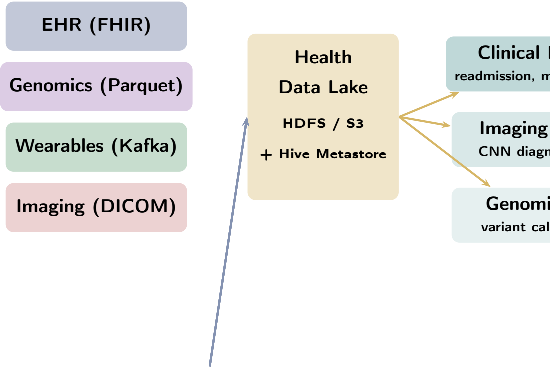
1. Load all SKU time series into a Spark DataFrame (one row per day per SKU)
2. Group by `sku_id`: each group is one independent series
3. Apply `applyInPandas(prophet_fit_and_forecast)`: Spark sends each SKU's data to one partition; Prophet runs inside a Pandas UDF
4. All 5,000 models fit in parallel across the cluster

Domain C: Healthcare Big Data

Healthcare Data Sources and Characteristics

Four Healthcare Data Modalities

1. **Electronic Health Records (EHR):** structured diagnoses (ICD-10), medications, lab results, vital signs, clinical notes (unstructured free text). Semi-structured: HL7 FHIR resources (Week 2).
2. **Genomics (Omics):** A single whole-genome sequence is ∼200 GB. A clinical genomics study of 10,000 patients = 2 PB. Requires specialised distributed storage (Apache Parquet + GATK pipeline on Spark).
3. **Wearables and IoT:** Continuous heart rate, SpO_2, sleep, glucose monitor readings. Generates time-series data at per-second resolution; requires streaming ingestion (Kafka) and anomaly detection.
4. **Medical Imaging (DICOM):** CT, MRI, X-ray,



Clinical Prediction Models and Privacy

EHR-Based Clinical Prediction (Obermeyer & Emanuel, 2016)

The landmark NEJM paper argued that **ML models trained on EHR big data** systematically outperform physician heuristics for three critical clinical tasks:

- **ICU mortality prediction:** Random forest trained on vitals, labs, and prior diagnoses achieves AUROC > 0.90 — identifying patients who will die within 24 hours earlier than conventional scoring (APACHE, SOFA)
- **30-day hospital readmission:** Gradient boosting on discharge records identifies high-risk patients for post-discharge follow-up, reducing readmission rates by 20%
- **Sepsis early warning:** Continuous streaming of

Privacy: HIPAA, GDPR, and Bangladesh

Regulation	Key Requirement
HIPAA (USA)	De-identify 18 PHI identifiers before research use; consent required
GDPR (EU)	Data minimisation; purpose limitation; right to erasure
PDPA Bangladesh	Under development; based on GDPR principles; NID-linked records require explicit consent

AlphaFold: Big Data + Deep Learning in Science

The 50-Year Protein Folding Problem

A protein's **3D structure** determines its function. From an amino acid sequence, predicting the 3D fold was considered one of biology's hardest problems. Experimental determination (X-ray crystallography) takes months per protein and costs thousands of dollars. **AlphaFold 2 (DeepMind, 2020):**

- Trained on the **Protein Data Bank**: 170,000 experimentally determined structures
- Architecture: **Evoformer** transformer block applied to multiple sequence alignments (MSAs) across evolutionarily related proteins
- Achieves median GDT score >92% (expert-level accuracy)
- **2022:** Released 200 million predicted protein

Why This Is a Big Data Problem

The input to AlphaFold is not just the query sequence — it is a **multiple sequence alignment (MSA)** of thousands of evolutionarily related sequences from millions of organism genomes:

- UniProt database: 250 million protein sequences (>200 GB compressed)
- MSA generation for one protein query: search 250 M sequences, align top hits, encode as a 2D matrix
- Training: required 128 TPU v3 cores for several weeks on the PDB + UniRef90

Bangladesh relevance: ICDDR,B and DGHS are using AlphaFold to study *Vibrio cholerae* protein structures relevant to cholera vaccine design — leveraging a globally trained model on locally

Case Study: CDC Syndromic Surveillance

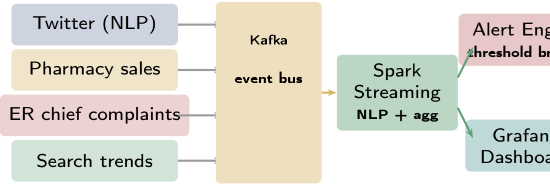
CDC Syndromic Surveillance: Big Data for Epidemic Early Warning

The Problem: Surveillance Lag

Traditional disease surveillance relies on **physician-reported cases** submitted to the CDC. The lag between a patient's first symptoms and the CDC receiving the report is **1–2 weeks**. By then, an outbreak may be exponentially growing. **Syndromic surveillance** uses **non-traditional data sources** that reflect illness *before* a diagnosis is made:

- **Social media:** Twitter/X posts containing words "flu", "fever", "can't taste" peak 7–10 days *before* ER visits
- **Pharmacy sales:** OTC antipyretic (paracetamol, ibuprofen) purchase spikes at 3–5 days before clinical confirmation
- **ER chief complaints:** text from triage notes

Syndromic Surveillance Pipeline



Lesson: Supplement, Not Replace

Google Flu Trends overfit to search behaviour and **over-predicted** flu incidence by 2× in 2013. Syndromic surveillance provides **early signals**, not ground truth — it must be combined with traditional clinical reporting.

Ivy League Alignment

Real-World Applications in Top University Curricula

Stanford CS246 — Mining Massive Datasets (Leskovec)

Stanford's application focus centres on **social network analysis** and **recommender systems**:

- **Influence maximisation** (Kempe et al., KDD 2003) is a core lecture topic. Students implement the greedy algorithm and compare with CELF (Cost-Effective Lazy Forward) optimisation on a 50-million-node Twitter graph
- **Community detection:** Leskovec's own work on network structure ("Communities in Networks", 2008) is assigned reading. Louvain and Spectral Clustering are compared on citation networks
- **Time series project:** Students forecast Wikipedia page view time series (2 billion rows) using Prophet and evaluate against ARIMA baselines — a real-world

Harvard CS265 — Big Data Systems (Idreos)

Harvard covers healthcare big data as the primary case study for **systems that must handle heterogeneous, privacy-sensitive data at scale**:

- The **UK Biobank** (500,000 patient genomics + EHR dataset) is the homework dataset — students run Spark jobs on 250 GB VCF files
- Differential privacy is implemented in a lab assignment: add Laplace noise to aggregate queries on patient cohorts

MIT 6.S897 — ML for Healthcare (Sontag)

Student Assessment Pool

Instructions

Select the single best answer. Correct answers **bolded** for instructor reference.

Q1. Facebook's Prophet library is primarily designed for:

- (a) Graph analytics and community detection in social networks
- (b) Real-time stream processing of events
- (c) **Time series forecasting with decomposable trend, seasonality, and holiday components**
- (d) Text sentiment classification using deep learning

Q2. Which algorithm is most commonly used for **community detection** in large-scale social network graphs?

- (a) K-Means clustering (b) PageRank (c) **Louvain algorithm** (d) ARIMA

Instructor Notes

Q1: Distractor (d) — Prophet has nothing to do with sentiment; it is a decomposable time series model. Distractor (a) — graph analytics uses GraphX/NetworkX, not Prophet. **Q2:** PageRank (b) measures node *importance*, not community membership. K-Means (a) clusters vectors, not graph topologies. ARIMA (d) is a time series method. Louvain (c) directly optimises modularity for community structure

Assessment—MCQ Set B *CLO4*

Q3. HIPAA in the United States primarily regulates the privacy and security of:

- (a) Financial transaction data and bank records
- (b) **Healthcare data including patient records and medical information**
- (c) Social media user data and behavioural profiles
- (d) Government data retention and public records

Q4. DeepMind's AlphaFold used big data and deep learning to solve:

- (a) Real-time financial fraud detection at bank-card transaction scale
- (b) Social network influence maximisation over billion-node graphs
- (c) **The protein structure prediction problem, predicting 3D structure from amino acid sequence**
- (d) Time series anomaly detection in IoT sensor streams

Instructor Notes

Q3: A common confusion — HIPAA (Health Insurance Portability and Accountability Act) is specifically for healthcare. Financial data is regulated by GLBA (US) or Basel III. **Q4:** AlphaFold's achievement was predicting protein 3D structure from sequence alone, using MSAs from 250 M sequences and training on the 170,000 experimentally solved structures in the Protein Data Bank. It won the 2024 Nobel Prize in Chemistry.

Instructions

State True or False and provide a **one-sentence justification** for full marks.

Statement	**Answer**
1. Google Flu Trends successfully replaced traditional CDC flu surveillance and has been used as the primary source ever since.	**False**
2. Time series data is order-sensitive: randomly shuffling the records in a time series changes the analysis result.	**True**
3. Social network graphs are typically sparse, with most nodes connected to only a small fraction of all possible neighbours.	**True**
4. Medical imaging data such as MRI and CT scans is classified as structured data because it is stored in files.	**False**

Assessment—Descriptive Questions *CLO4*

Instructions

Answer each question in **100–150 words**.

D1. Describe **three types of analysis** commonly performed on social media data (e.g., community detection, sentiment analysis, influence maximisation). For each, state one specific **ethical risk** that arises when the analysis is deployed at scale.

D2. What is the difference between **ARIMA** and **Facebook Prophet** for time series forecasting? Under what conditions would you prefer Prophet over ARIMA? When might ARIMA still be the better choice?

D3. Identify **three data integration and privacy challenges** specific to healthcare big data analytics. For each, describe one technical mitigation strategy (e.g., differential privacy, federated learning, de-identification).

Instructions

Answer in **200–300 words** with end-to-end pipeline reasoning.

A1. Flu Outbreak Early-Warning System for Bangladesh:
Design a big data system to detect flu outbreaks in Bangladesh using Twitter/Facebook posts and pharmacy purchase records from major pharmacy chains.

- (a) How would you **collect and ingest** these two data sources? (Specify tools: Kafka, Sqoop, Flume, REST API)
- (b) What **features** would you extract from each source for a prediction model? (Name specific NLP features and pharmacy signal metrics)
- (c) What **model** would you use, and how would you validate it against ground-truth CDC/DGHS surveillance data?
- (d) What are the **ethical risks** of this system? How would you mitigate them?

A2. Retail Demand Forecasting Pipeline:
A large Bangladeshi supermarket chain needs 90-day demand forecasts for 5,000 store–product SKU combinations. Design the **end-to-end big data pipeline**: (a) data ingestion from POS systems; (b) storage and partitioning strategy in HDFS/S3; (c) forecasting model choice (ARIMA

Textbook & Reading References

Primary Textbooks for Week 10

- **[SA]** Acharya & Chellappan. *Big Data and Analytics*, Wiley, 2015. **Ch. 11**
  - Ch. 11: Big Data Use Cases — social media, healthcare, retail, government analytics
- **[TE]** Erl, Khattak & Buhler. *Big Data Fundamentals*, Prentice Hall, 2016. **Ch. 13**
  - Ch. 13: Big Data in Practice — domain-specific deployments and lessons learned
- **[LRU]** Leskovec, Rajaraman & Ullman. *Mining of Massive Datasets*, 3rd ed. **Ch. 10**
  - Ch. 10: Social-Network Graphs — community structure, influence propagation, PageRank

Seminal Papers (covered in Week 10, Part 2)

- Kwak, H. et al. (2010). **What is Twitter, a Social Network or a News Media?** *WWW 2010.*
- Taylor, S. J., & Letham, B. (2018). **Forecasting at Scale.** *The American Statistician.* [Facebook Prophet paper]
- Obermeyer, Z., & Emanuel, E. J. (2016). **Predicting the Future — Big Data, Machine Learning, and Clinical Medicine.** *New England Journal of Medicine.*
- Jumper, J. et al. (DeepMind, 2021). **Highly Accurate Protein Structure Prediction with AlphaFold.**

Questions & Discussion

Week 10, Part 1 | CSE 4345 Big Data Analytics

"Big data is not about the data. It is about the decisions."

— Kenneth Cukier & Viktor Mayer-Schönberger, *Big Data: A Revolution*, 2013

Next Lecture: Week 10, Part 2

Seminal Papers: Kwak et al. (2010) Twitter as news medium · Taylor & Letham (2018) Prophet · Obermeyer & Emanuel (2016) clinical ML

Case Study: **CDC Syndromic Surveillance** — integrating Twitter, pharmacy, and ER data to detect flu outbreaks 1–2 weeks before clinical confirmation

Reading before Part 2: [LRU] Ch. 10 · Taylor & Letham (2018) Prophet paper (open access)